

INTERACTIVE 3D SOLAR SYSTEM LEARNING PLATFORM

ASTRA

Explore. Visualize. Understand the Solar System.



The Solar System is 3D. Why learn it in 2D?



Traditional Methods

- ✗ Static textbooks and 2D diagrams.
- ✗ Zero scale comprehension.
- ✗ Passive video consumption.

Read → Memorize → Move On



The ASTRA Method

- ✓ Spatial interactive exploration.
- ✓ True planetary relationships.
- ✓ Contextual AI guidance.

Explore → Interact → Understand



One Interactive Space for Understanding

ASTRA turns passive astronomy learning into active exploration.



Explore

Interact with planets in realistic 3D space.



Understand

Visualize astronomical data dynamically.



Compare

Place different worlds side by side.



Experiment

Change physics and observe model scenarios.



Learn

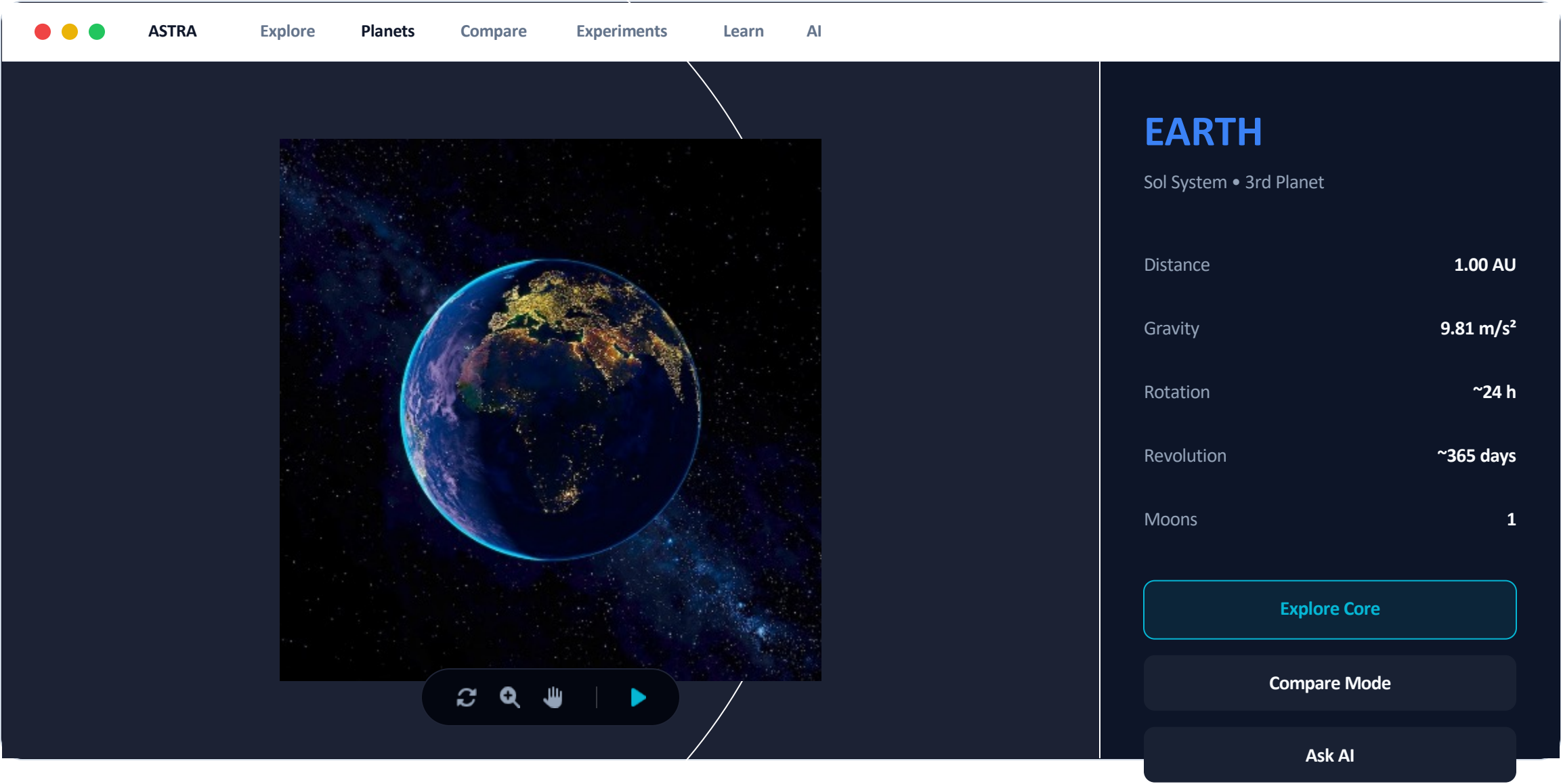
Grasp concepts through pure visualization.



Ask AI

Receive contextual educational explanations.

What Will Our Website Look Like?



EARTH

Sol System • 3rd Planet

Distance	1.00 AU
Gravity	9.81 m/s ²
Rotation	~24 h
Revolution	~365 days
Moons	1

Explore Core

Compare Mode

Ask AI

Click a Planet. Understand Its World.

[Back to Solar System](#)

MARS

The Red Planet

PHYSICAL

ATMOS

MOONS

DISTANCE FROM SUN

~1.52 AU

GRAVITY

~3.71 m/s²

ROTATION

~24.6 h

REVOLUTION

~687 days

ATMOSPHERE

Thin, CO₂-rich

AVERAGE TEMPERATURE

~-63°C

Compare

Ask AI

How ASTRA Helps Students Learn

ASTRA makes astronomy an experience, not just information.



MEMORIZE → VISUALIZE → INTERACT → QUESTION → UNDERSTAND

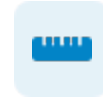
How Big? How Far?

Visual scales may be adjusted for learning clarity while scientific values remain available.



Orbit View

Solar System structure and paths.



Distance View

Relative distances from the Sun.



Size View

Compare pure planetary dimensions.



Educational View

Adjusted scales for clear learning.

Rotation ≠ Revolution



ROTATION

Earth spins on its axis.

~24 hours



REVOLUTION

Earth completes an orbit.

~365 days

MERCURY ORBIT SPEED
~47.4 km/s

EARTH ORBIT SPEED
~29.8 km/s

NEPTUNE ORBIT SPEED
~5.4 km/s

Compare Worlds Side by Side

Back to Explore

COMPARISON MODE

View Options

EARTH



Gravity

9.81 m/s²

Atmosphere

N₂, O₂

Moons

1

MARS



Gravity

3.71 m/s²

Atmosphere

Thin, CO₂

Moons

2

AI QUESTIONS

Why is Mars red?

Why is Earth's gravity stronger?

Why is Mars colder?

Ask. Change. Simulate. Understand.

Advanced educational simulations based on model outcomes.

EXPERIMENT PARAMETERS

Gravity Multiplier +20%



Orbital Distance 1.00 AU



RUN EXPERIMENT

OTHER SCENARIOS:

What if Earth had no Moon?

What if Earth stopped rotating?

Scenario: Earth Gravity +20%

ILLUSTRATIVE SCIENTIFIC SCENARIO

CURRENT EARTH

9.81 m/s²

MODIFIED SCENARIO

11.77 m/s²

AI MODEL EXPLANATION

Model-based educational outcome: Increasing Earth's mass to achieve 20% more gravity would significantly impact atmospheric pressure. The atmosphere would become denser, trapping more heat. Surface water distribution would alter, and escape velocity would increase from 11.2 km/s to ~12.2 km/s, making space travel much more difficult.

Ask. Change. Simulate. Understand.

ASTRA

Explore

Planets

Compare

Experiments

Learn

AI

Experiment Matrix

ACTIVE TARGET

●

Earth

PARAMETERS

Gravity+20%

Illustrative scientific scenario. Model-based educational outcome.



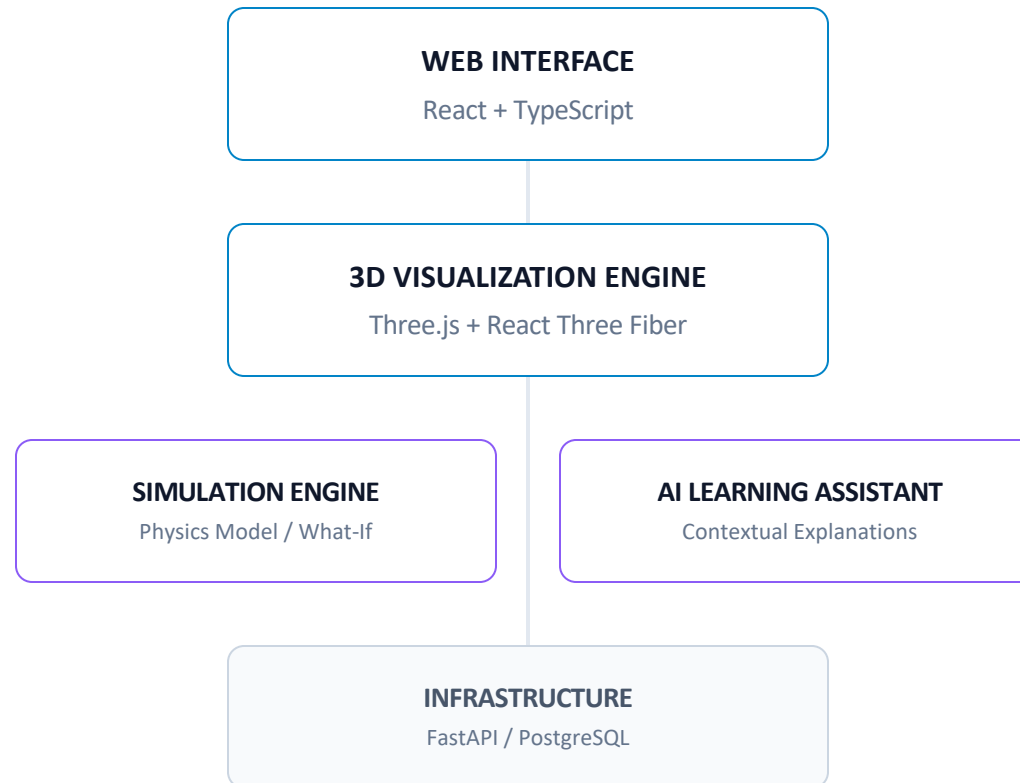
AI ANALYSIS

"Here's what the model suggests:"

- Denser atmospheric pressure.
- Altered weather patterns.
- Increased surface escape velocity.

How ASTRA Works

A modern web stack for 3D simulation and AI learning.



From Website to Digital Planetarium

ASTRA evolves from an educational tool into an immersive platform.

PHASE 01: TODAY

Interactive 3D Solar System

- Core planet exploration.
- Distance and size scaling.
- Basic data visualization.

PHASE 02: NEXT

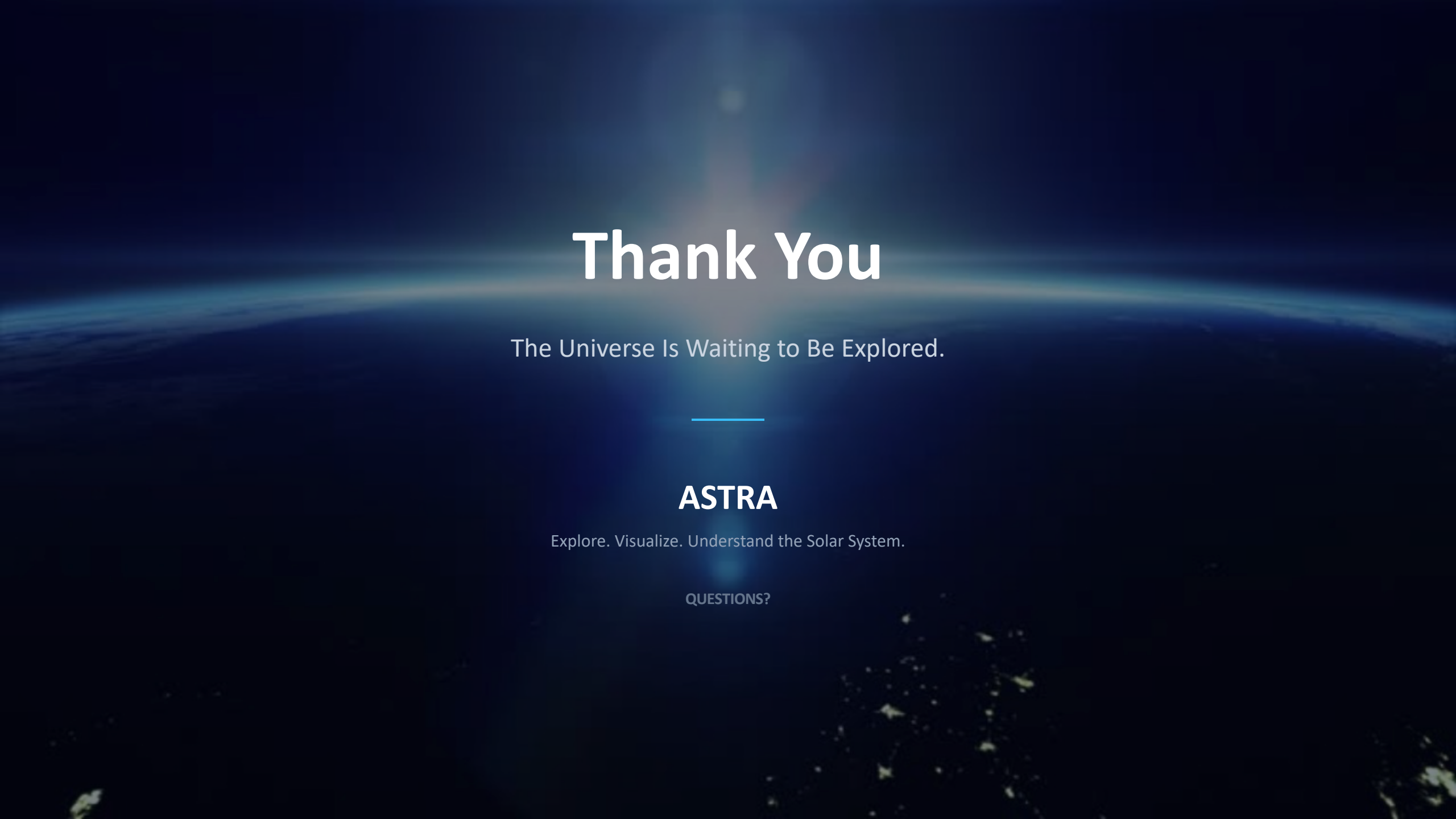
Advanced Simulation

- Scientific What-If engine.
- Exoplanet exploration.
- AI Personalized learning.

PHASE 03: FUTURE

Digital Planetarium

- VR / AR Astronomy.
- Classroom integrations.
- Immersive group learning.



Thank You

The Universe Is Waiting to Be Explored.

ASTRA

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QUESTIONS?